

A Comparative Study of Traditional and Modern Approaches to Portfolio Construction and Performance Evaluation

Does portfolio optimization outperform naïve diversification?

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[°] Represents aspiration, and not current job role

“In the stock market the more elaborate and abstruse the mathematics the more uncertain and speculative are the conclusions we draw there-from.”

- Ben Graham, “Father” of Value Investing

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Section 1: Introduction

Imagine, if you will, a basket filled with mangos, bananas, oranges, and lemons. What is the most “optimal” composition of this basket? There is no one right answer, or is there? To understand the right composition, you must first understand your desires. If you wish for a sweeter experience, the proportion of mangos, and bananas will be more than that of the lemons (or you may remove the lemons entirely!). If you wish for a balanced experience, you might keep all the fruits in the same proportions. Or, if you want a more “citrusy” experience, you might increase the proportions of oranges, and lemons, with respect to mangos and bananas. To “optimize” your experience, you must strike a balance between what you want, *versus* what you are willing to give up. Now, replace the fruits with businesses, and “basket”, with “portfolio”, and you will have intuitively developed some understanding of portfolio management theory!

Portfolio management, in the simplest of terms, can be defined as an exercise of choice. In formal terms, portfolio management is the “art” of selecting and overseeing a group of investments that meet the long-term financial objectives and risk tolerance of a client, a company, or an institution. The field of portfolio management has evolved, from humble beginnings, where decisions were arbitrarily taken, to modern techniques, which employ advanced mathematics, data science, statistics, and even physics. Though the methodology for portfolio management has changed drastically, the underlying goal has been the same, which is to maximise returns for a given level of risk, or to minimise risk. Though various methods exist, all of them intend to optimize diversification, to strike the optimal risk reward ratio.

To understand optimal diversification, we employ an example popularised by Burton G. Malkiel, in his book, *A Random Walk Down Wall Street*:

Consider a small island economy with two businesses, a resort hotel and an umbrella manufacturer. The performance of the resort depends on sunny weather, while the umbrella manufacturer performs better during rainy conditions. An investor who allocates capital solely to one of these businesses faces significant risk if weather conditions are unfavorable. However, by investing in both enterprises, the investor benefits from the offsetting nature of their returns. When one performs poorly, the other is likely to perform well, thereby stabilizing overall portfolio outcomes.

This simple illustration captures the fundamental principle of optimal diversification: combining assets with differing responses to the same external conditions can reduce overall portfolio risk. With this example, we transition to understanding the purpose of this study. While the principle of diversification is widely accepted, the methods used to construct diversified portfolios vary significantly. Some investors rely on simple heuristic allocation strategies, while others employ mathematically optimized models derived from modern portfolio theory. This research addresses that gap by subjecting traditional equal-weight heuristics and mean-variance optimised portfolios to identical empirical testing, using a shared dataset and a common set of risk-adjusted performance metrics. The study thereby determines which approach yields superior results and whether the theoretical advantages of optimisation are preserved under realistic investment constraints.

The next section explores the origins of portfolio theory, and how it evolved, after which the research framework, methodology, and portfolio construction are detailed, along with the testing of the research hypotheses (detailed in section 3.3). The penultimate section and ultimate section discuss the viability of traditional and optimized strategies, and suggest actions to investors, while evaluating the limitations of this study, and the modern portfolio theory.

Section 2: Literature Review

The history of portfolio optimization techniques is rich, and traces its roots back to a paper authored by Harry Markowitz in 1952, aptly titled *Portfolio Selection*, where he detailed how assets shouldn't be evaluated in isolation, but with a "bird's eye" view of the risk-return levels of the entire portfolio. Markowitz with his ground-breaking work, introduced a mean-variance, and the efficient frontier. Then in 1958, James Tobin built on Markowitz's work, and introduced the concepts like that of the "risk-free asset", the two fund separation theorem, and the capital market line (CML), in his paper *Liquidity Preference as a Behavior Toward Risk*. Prominent economists Sharpe & Lintner (1964/65)¹, brought to the world, the concepts of the Capital Asset Pricing Model (CAPM), and metrics such as the *beta* which was a measure of risk. Later works turned inwards. Respective works by Michaud (1989), and Best & Grauer (1991) "ruminated" on how small errors in expected return inputs caused wildly unstable, concentrated portfolio weights, and termed it the *estimation error problem* (spoiler alert! This work demonstrates this problem).

In 2009, DeMiguel, Garlappi & Uppal, released a critical paper, titled, *Optimal versus Naive Diversification: How Inefficient Is the 1/N Portfolio Strategy?* Which illustrated that the "1/N rule"², outperforms a sophisticated optimization across most datasets (we can't help mention a quote by John C. Bogle, which he wrote in his rightly titled book *Common Sense Investing*, "don't buy the needle, buy the haystack". Though this quote about active and passive funds, the parallels are evident!). Later, in 2012, Plyakha, Uppal & Vilkov, in their paper titled, *Why Does an Equal-Weighted Portfolio Outperform Value- and Price-Weighted Portfolios?* Detailed how equal-weight strategies, by inherently enforcing the "buy low, sell high" rebalancing methodology, partly outperform other strategies over the long run³. Perhaps the most ground breaking paper on this matter was authored decades before, in 1973. In their work *Risk, Return, and Equilibrium: Empirical Tests*, Fama & MacBeth provided for a groundwork backed by empirical testing.

Portfolio optimisation literature has also seen works with a behavioural angle. Benartzi & Thaler (2001) detailed how people naturally gravitate to 1/N in retirement allocation. Naïve diversification isn't just academically competitive, it's psychologically intuitive. Passive management is the practical expression of CAPM, hold the market, minimise costs. It is important to mention John Bogle's Vanguard here, who brought these benefits to the masses through the launch of index funds.⁴

Before we progress, we must ask the ever-important question which has guided man through millennia. The question of *why*. Why write this paper? Why add to the giant mountain of literature? Most studies either test many complex optimisation models against each other, or discuss active/passive broadly. We, by this work, set out to explore, through a direct empirical comparison, using the same asset universe, the same metrics, the merits of the equal-weight strategy vs the mean-variance, especially within a constrained retail-style portfolio.

1 Sharpe (1964), Lintner (1965)

2 The 1/N rule is a simple investment strategy where an investor divides their total investment equally among N different assets or funds. This approach is often used for diversification, as it helps to spread risk without relying on complex calculations.

3 By periodically rebalancing to equal weights, the strategy automatically sells recent winners and buys recent losers, systematically capturing mean reversion and enforcing a "buy low, sell high" discipline.

4 Warren Buffett, perhaps the most prolific active managers, on Bogle: *Jack (John) Bogle has probably done more for the American investor than any individual I've known*

Section 3: Research Framework

3.1 Research Objectives

This research intends to examine the effectiveness of different portfolio construction strategies in achieving optimal risk-reward trade-offs. It analyses the impact of diversification and asset correlation on portfolio risk. This work evaluates the role of optimisation techniques, particularly mean-variance analysis, in improving portfolio efficiency. Finally, this study compares the performance of a traditional (equal-weight) and a modern (optimized) portfolio, representing passive and active strategies respectively^o.

3.2 Research Questions

- How do active and passive portfolio strategies differ in terms of portfolio returns?
- To what extent do diversification and asset correlation influence portfolio risk?
- How do systematic and unsystematic risk factors affect risk-adjusted portfolio performance?
- Does the use of portfolio optimization techniques improve portfolio efficiency?
- Do modern portfolio construction approaches generate superior risk-adjusted performance compared to traditional methods?

3.3 Hypotheses of the Study

H1: Active and passive portfolio construction strategies differ in their impact on portfolio returns.

H2: The level of diversification and asset correlation significantly influence portfolio risk, measured through standard deviation and beta.

H3: Risk factors including systematic and unsystematic risk significantly affect risk adjusted returns, measured through Sharpe Ratio and Jensen's Alpha.

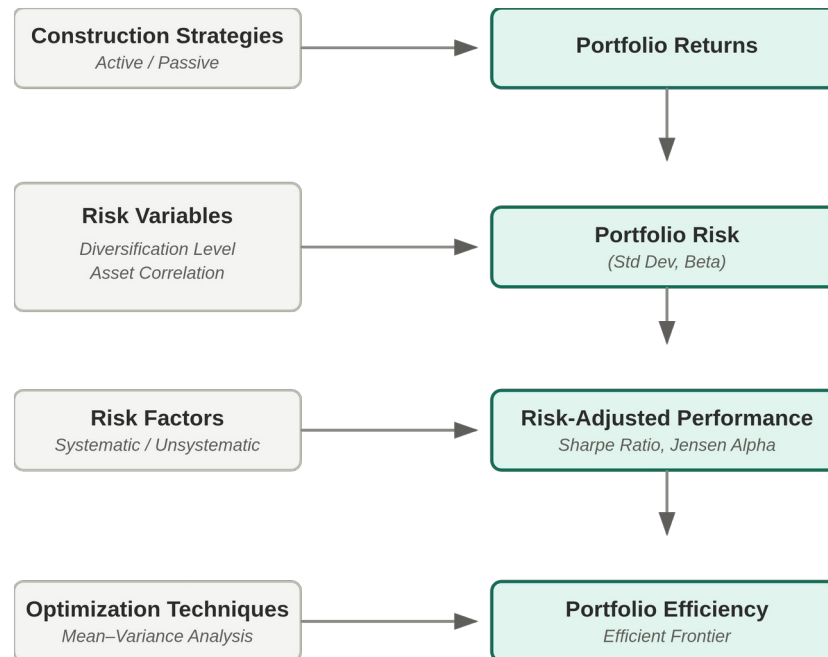
H4: The use of optimization techniques such as mean variance analysis leads to higher portfolio efficiency and improved

positioning on the efficient frontier.

H5: Modern portfolio construction methods combined with active or passive strategies are expected to generate superior risk adjusted performance compared to traditional approaches

^o This study assumes that passively managed portfolios follow an equal-weight allocation. While this assumption simplifies the analytical framework, it does not fully reflect real-world passive investment strategies, which are often market-capitalization weighted. This limitation is discussed further in Section 7.2.

3.4 Conceptual Framework of the Study



Section 4: Research Methodology

4.1 Research Design

The design of this paper employs a quantitative approach to assess the performance of the traditional way of portfolio construction versus modern optimization methods. This quantitative approach allows accurate and precise measurement of key metrics, such as risk, return, and risk-adjusted performance. This study uses historical data (1st January 2019 - 6 March 2026), and zooms in on Reliance Industries (RELI) (Energy / Conglomerate), HDFC Bank (HDBK) (Banking), Eicher Motors (EICH) (Automobile), Hindustan Unilever (HLL) (FMCG), Bharti Airtel (BRTI) (Telecom), Larsen & Toubro (LART) (Infrastructure / Engineering), Sun Pharmaceutical Industries (SUN) (Pharmaceuticals), Titan Company (TITN) (Consumer Discretionary), Tata Steel (TISC) (Metals)[◦]. These companies were selected because this portfolio is diversified across sectors. These companies have large-cap stability, and hold leadership positions in their markets. These companies differ in volatility and beta, which is optimal for studying diversification, and other metrics. These companies also have extensive coverage, which ensured data availability. These companies have significant retail and institutional interest, and hence represent a good pool to assess under different market conditions. Due to the limited sample of equities included in the study, a proxy market portfolio (henceforth referred to as “the benchmark”) was constructed using the equally weighted average returns of the selected securities. This synthetic market return was used for the estimation of CAPM beta and Jensen’s Alpha. The synthetic benchmark does not represent the broader market but serves as a controlled proxy within the study’s restricted asset universe. The risk free rate was set at 5% p.a. This rate was chosen for model simplification, even though a 10-year G-Sec yield is

[◦] Ticker symbols reflect the naming convention used by [investing.com](https://www.investing.com) and may differ from the official NSE symbols. This is a deliberate and conscious step to preserve study reproducibility, and preserve the original datasets.

typically closer to 6.5%–7.5%. Using a constant risk-free rate is a limitation, as in reality, it fluctuates and would affect the optimal portfolio calculation over the 7-year period.

4.2 Data Collection

This study is based solely on secondary data, because the “basket of companies” this paper studies is already extensively covered, and the analysis uses monthly share movements. The time frame of 1st January 2019 - 6 March 2026 was chosen because it ensures coverage pre, during, and post pandemic along with geopolitical, and trade tensions. Python (libraries like pandas, and numpy, seaborn) was primarily used to clean and assess, and visualize the data. The entire project is open sourced and can be found [here](#) (the hyper-link can also be found in the references section). All data was sourced from publicly available sources, and no confidential/private data was accessed.

4.3 Sample Selection

As mentioned in section 4.1, this study focuses on a set of large-cap Indian companies, which spans diverse sections of the economy. These companies represent sectors like energy, banking, automobiles, FMCG, telecom, infrastructure, pharmaceuticals, consumer discretionary, and metals. This ensures proper diversification. These companies show varying levels of price sensitivity, and hence are important for assessing risk-return dynamics and the effects of diversification. Due to the limited sample of equities included in the study, a proxy market portfolio was constructed using the equally weighted average returns of the selected securities. This synthetic market return was used for the estimation of CAPM beta and Jensen’s Alpha.

4.4 Portfolio Construction

Asset	Equal_Weight (%)	Optimized_Weight (%)
RELI	11.11	0
HDBK	11.11	0
EICH	11.11	10.38
HLL	11.11	0
BRTI	11.11	52.16
LART	11.11	0
SUN	11.11	16.02
TITN	11.11	17.24
TISC	11.11	4.2

Table 4.4: Allocation

This study compares two approaches to portfolio construction: traditional equal-weighted portfolios and modern optimization-based portfolios. In traditional equal-weighted portfolios, all “scrips” (companies) are given the same allocation. As shown in “Equal_Weight(%)” (*Table 4.4*), all the scrips get an equal allocation of 11%. This is a simple and intuitive way to allocate capital, hence

serving as an apt baseline for further probing. Another advantage of this approach is that it helps test the effect of absolute diversification on risk and return.

The optimized (or modern) portfolio involves employing various techniques (or “strategies”) to maximise returns for a given risk. This approach considers volatility, beta, and correlations. The allocations in “Optimized_Weight (%)” (*Table 4.4*) are based on mean-variance optimization (Markowitz Modern Portfolio Theory (MPT)). In simple terms, the model (which can be found [here](#), and in the references) employed expected returns, variances, and covariances of the assets to determine optimal weights. The goal was to either maximise the “Sharpe Ratio” (risk-adjusted returns) or minimize portfolio variance (volatility). This portfolio does not employ any short selling mechanisms, i.e., the weight of each scrip is always greater than, or equal to zero. This is a deliberate choice taken because most retail and institutional investors in India do not engage in short-selling . Multiple portfolios were constructed and evaluated across risk-return profiles, to identify the efficient frontier (the combination which yields the “best” risk-reward ratio). The final optimized portfolio corresponds to the point on the frontier with highest Sharpe ratio, which explains why some assets have much higher weights than others (RELI having 0% weight, while Airtel has 52.16% weight). This allocation might appear “radical” at first, but this is precisely the difference between the intuitive traditional, and mathematical modern approach to capital allocation. The optimized portfolio is “crafted” with the goal of maximising portfolio efficiency. Optimized portfolios are sensitive to input assumptions (expected returns, covariance estimates), so results can vary with different data or models

It is important to note that this portfolio is purely theoretical and does not take into account any risk management norms (as illustrated by Airtel having a 52% weight). In real world approaches weight caps are in place to prevent over exposure (as in this case). The model favoured Airtel to such a vast extent because of its low correlation with other assets in the portfolio. On a sectoral level, the telecom sector is a growing sector, powered both by policy, and private interest. Firms enjoy relatively defensive cashflows (broadly supported by repeat customers) , which make them reliable investments. These market structure advantages provide for a systemic reason for such high allocation.

4.5 Analytical and Statistical Techniques

Expected Returns

The model calculated the average historical returns of each asset to estimate future performance.

$$E(R_i) = \frac{1}{T} \sum_{t=1}^T R_{i,t}$$

Where $E(R_i)$ is the expected return of asset i , $R_{i,t}$ is the return at time t , and T is the total number of periods. This provides a baseline expectation of performance for portfolio construction.

Risk Assessment (Volatility & Variance)

The model measured individual asset risk using variance and standard deviation of historical returns.

Variance:

$$\sigma_i^2 = \frac{1}{T-1} \sum_{t=1}^T (R_{i,t} - E(R_i))^2$$

Standard Deviation:

$$\sigma_i = \sqrt{\sigma_i^2}$$

Where σ_i^2 is the variance, and σ_i is the standard deviation (volatility). This “captures” individual asset risk and informs optimized allocation.

Covariance & Correlation

The model examined how asset returns move together to assess diversification using:

Covariance:

$$Cov(R_i, R_j) = \frac{1}{T-1} \sum_{t=1}^T (R_{i,t} - E(R_i))(R_{j,t} - E(R_j))$$

Correlation:

$$\rho_{i,j} = \frac{Cov(R_i, R_j)}{\sigma_i \sigma_j}$$

Where $\rho_{i,j}$ ranges from -1 (perfect negative correlation) to +1 (perfect positive correlation). This determines benefits of combining assets with differing risk-return behaviour.

Portfolio Optimization (Mean-Variance / MPT)

The model constructed portfolios to maximize risk-adjusted return (Sharpe ratio) under constraints: $\sum w_i = 100\%$, $w_i \geq 0$. (w_i = weight of asset i in the portfolio)

Sharpe Ratio (Risk Adjusted Returns)

$$S_p = \frac{E(R_p) - R_f}{\sigma_p}$$

Portfolio Variance:

$$\sigma_p^2 = \sum_{i=1}^N \sum_{j=1}^N w_i w_j Cov(R_i, R_j)$$

Where, S_p = Sharpe ratio of the portfolio, R_f = risk-free rate (e.g., government bond yield), $E(R_p)$ = expected portfolio return, σ_p = standard deviation (volatility) of the portfolio. This identifies the efficient frontier, selects the portfolio with the highest Sharpe ratio, and maximizes portfolio efficiency.

Synthetic Market° Formula:

A proxy market portfolio was constructed using the equally weighted average returns of the nine sample equities. This synthetic market return was used as the market factor for estimating beta and Jensen's Alpha within the CAPM framework.

$$R_m = \frac{1}{N} \sum_{i=1}^N R_i$$

Where R_m represents the return of the synthetic market portfolio, R_i denotes the return of the i^{th} security, N represents the total number of securities in the portfolio, and i is the index of each security included in the summation.

Performance Metrics

Portfolios were evaluated relative to a synthetic benchmark using the following:

Beta (β_p): Measures systematic risk of the portfolio relative to the market

$$\beta_p = \frac{\text{Cov}(R_p, R_m)}{\sigma_m^2}$$

Where, R_p = portfolio return, R_m = benchmark return (synthetic), σ_m^2 = variance of market returns, and $\text{Cov}(R_p, R_m)$ = covariance between portfolio and market returns.

Jensen's Alpha (α_p): Measures excess return over CAPM expectation

$$\alpha_p = R_p - [R_f + \beta_p (R_m - R_f)]$$

Where, R_p = portfolio return, R_f = risk-free rate, β_p = portfolio beta, and R_m = market return.

Treynor Ratio (T_p): Return per unit of systematic risk

$$T_p = \frac{R_p - R_f}{\beta_p}$$

Where T_p = Treynor ratio of the portfolio, R_p = portfolio return, R_f = risk-free rate, and β_p = portfolio beta.

These metrics provide a risk-adjusted performance evaluation and allows direct comparison across portfolio approaches.

4.6 Performance Evaluation

° The proxy market portfolio represents the mean return of the selected securities and serves as an approximation of the market portfolio within the restricted asset universe of the study.

Portfolio	Sharpe	Beta	Treynor	Jensen_Alpha
Equal Weight	0.3351721565	1.0119047619	0.0161440354	-0.0001944788
Optimized	0.40965025	0.9122492503	0.02416554970	0.007142294360

Table 4.6(a): Portfolio Metrics

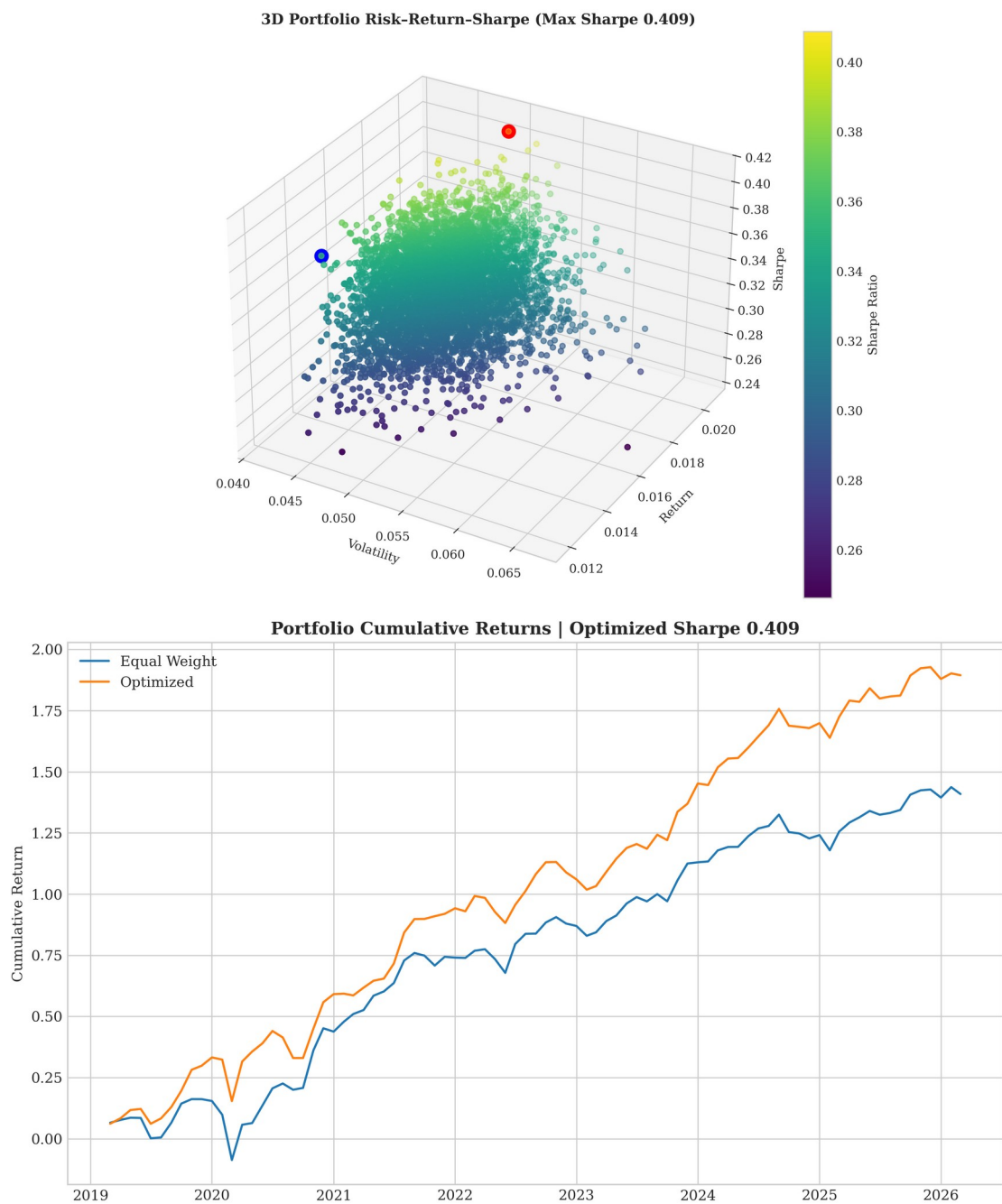


Chart 4.6 (b)

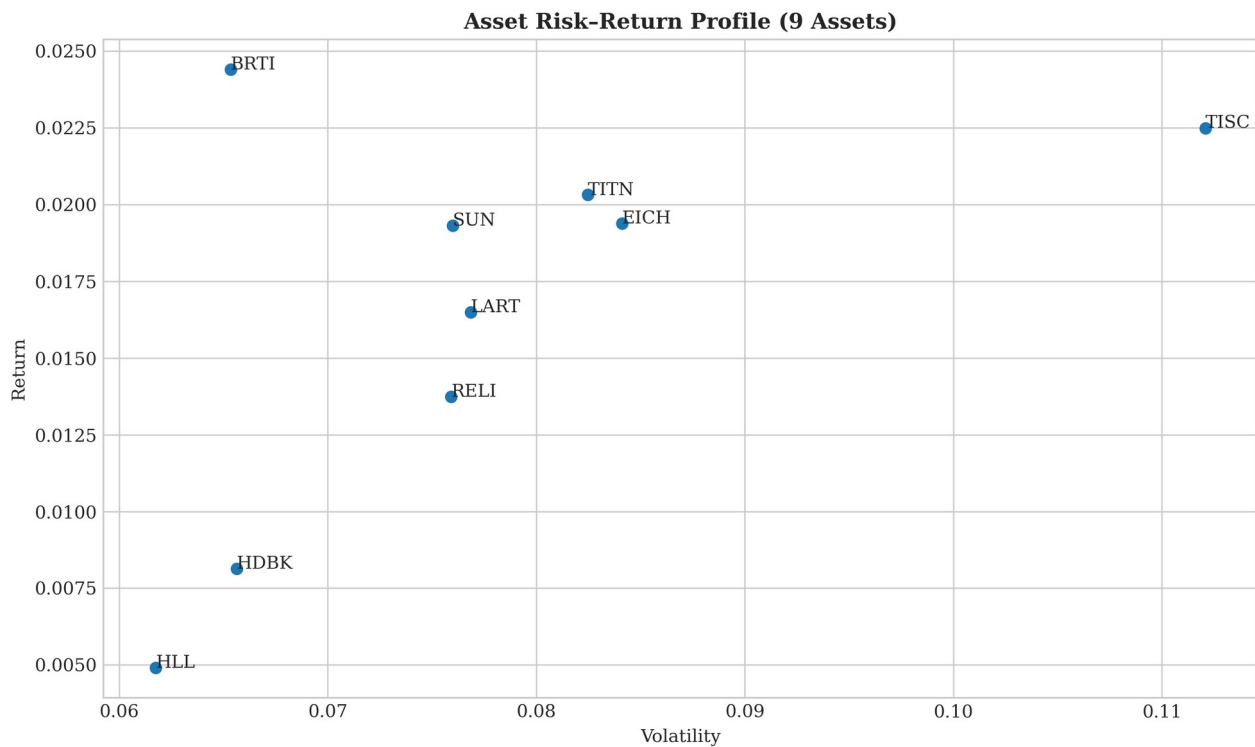


Chart 4.6 (c)

On analysis, both methods (traditional and modern) yielded vastly different results. As demonstrated in *chart 4.6 (a)*, across the multi-portfolio (as shown by the scattered dots) simulations, the efficient frontier was found where the Sharpe ratio was 0.409, indicating the most optimal risk-reward ratio. The Sharpe ratio (as shown in *table 4.6 (a)*) for the optimized portfolio (0.409) was higher compared to the equal weighted portfolio (0.335), which shows that the optimized portfolio had better risk-adjusted returns. The Beta (*table 4.6(a)*) for the optimised portfolio (0.91) was lower than the equal-weighted portfolio, demonstrating lower sensitivity in normal-risk evaluation. The Treynor ratio (*table 4.6(a)*) for the optimized portfolio (0.024) was higher than the Treynor ratio for the equal-weighted portfolio (0.016), showing a higher return per unit of systemic risk for the optimized portfolio. Jensen's Alpha (*table 4.6(a)*) for the optimized portfolio was higher compared to the equal weighted portfolio, showing a clear outperformance. Adjusted weights in the optimized portfolio reduced exposure to higher-risk assets while leveraging returns from stable performers.

As illustrated in *chart 4.6 (b)*, in the period between 2019 and 2026, the optimized portfolio delivered a [cumulative] return of almost 20%, whereas the equal weighted portfolio fell just shy of 15%, showing that the optimized portfolio outperformed the equal weighted portfolio by roughly 500 basis points.

This shows how modern optimization enhances efficiency, manages risk, leverages diversification, improves reward per risk.

Section 5: Data Analysis & Portfolio Construction

5.1 Descriptive Statistics of Asset Returns

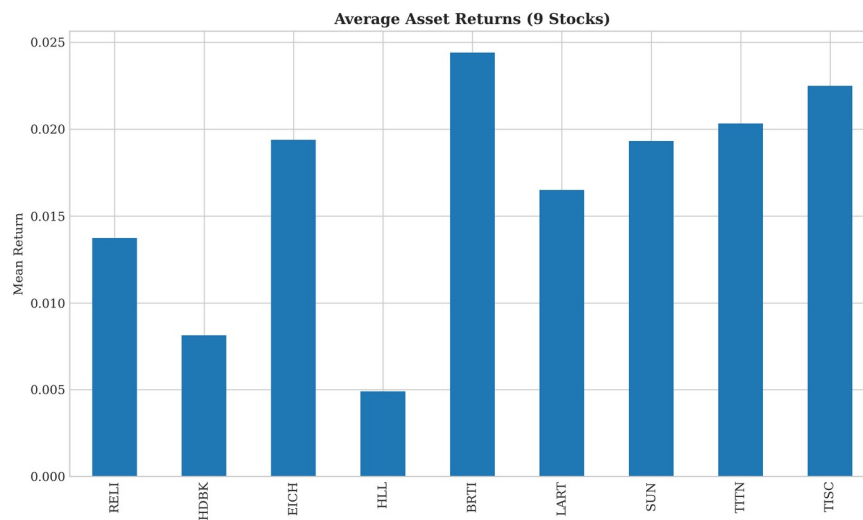


Chart 5.1 (a)

The monthly return data reveal considerable variation across sectors, which indicates a heterogeneity (diversity) in risk-reward behaviour. The sample period includes three unique phases, pre-pandemic stability (2019), the COVID pandemic panic (2020), and the recovery post the pandemic (2021-2026). This mixed environment allows one to observe “asset resilience” under extreme volatility conditions. Certain assets from the stock basket created for this paper demonstrated higher average returns (*charts 5.1 (a) and 5.1 (b)*) reflecting stronger long-term growth expectations. Sectors such as telecom displayed a stronger return momentum compared to sectors like FMCG which showed moderate (but stable) average returns.

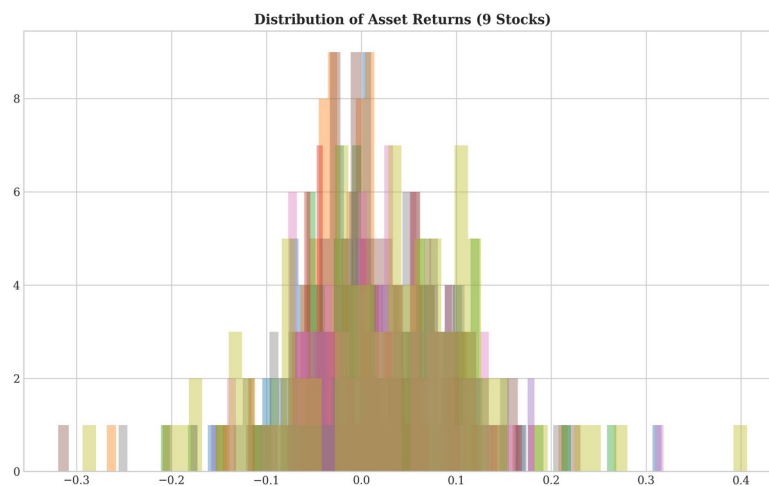


Chart 5.1 (b)

A look at volatility patterns (*chart 4.6(c)*) reveals that cyclical sectors such as metals and automobiles exhibit significantly higher standard deviations, indicating greater impact of macroeconomic changes. Defensive sectors (like FMCG’s Hindustan Unilever) demonstrated lower volatility, supporting their role as stabilizers in diversified portfolios. Data also revealed that

minimum returns across several assets coincide with the pandemic-driven market correction in March 2020.

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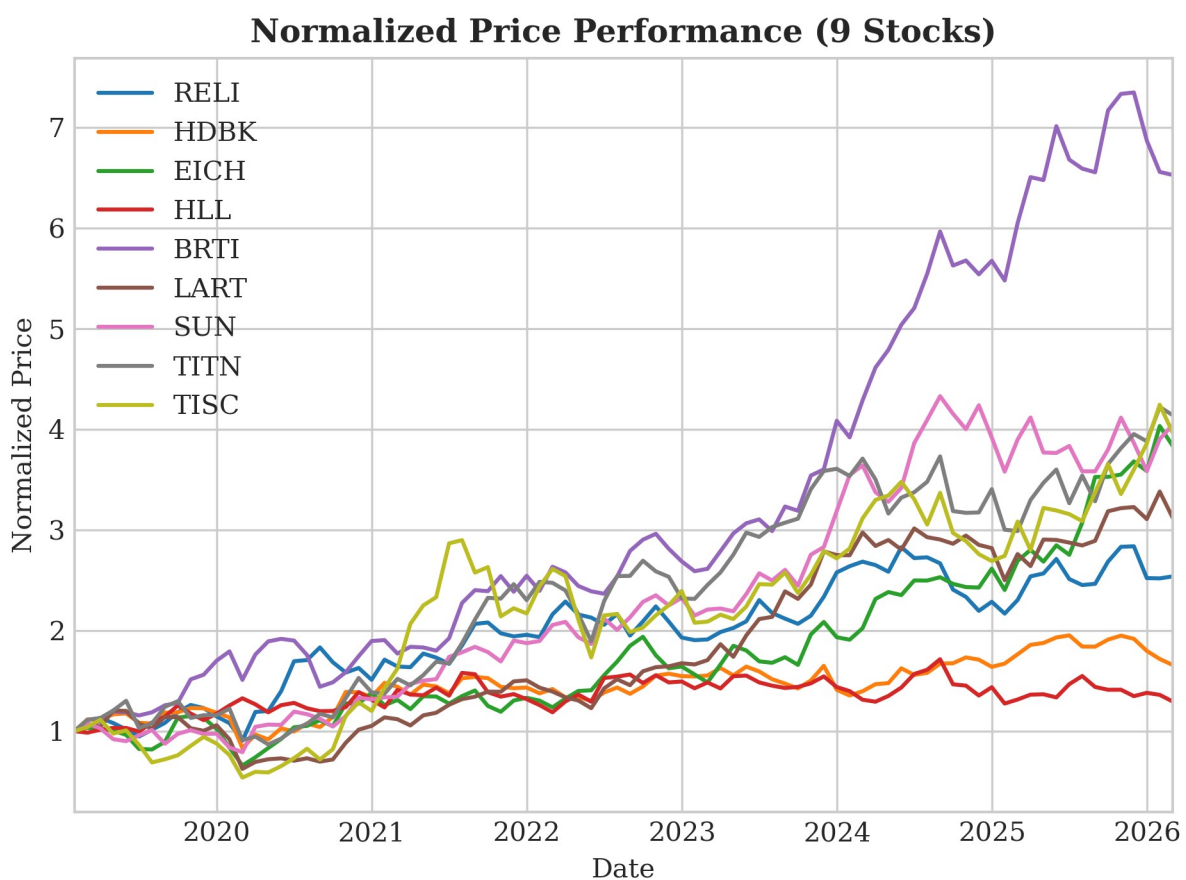


Chart 5.1 (c)

As shown in *chart 5.1 (c)*, some assets also showed large positive “rebounds”, highlighting the rapid recovery phase in Indian equities during late 2020 and 2021. These trends, along with the mean returns shown in *chart 5.1 (a)* demonstrate the importance of diversification in managing downside risk. Return distribution (*chart 5.1 (b)*) for several assets appear non-symmetric (or non-uniform), which shows variation in individual asset return behaviour. Some assets exhibited fat-tailed behaviour, indicating higher probability of extreme returns. This “small” detail plays a huge role in portfolio strategies as, traditional portfolios [often] assume normally distributed returns. These statistics reveal an important insight. Higher volatility does not necessarily translate into

proportionally higher returns. On the other hand, the variation in risk-return characteristics provide a strong foundation for diversification benefits. These differences justify the use of modern portfolio optimization techniques to construct efficient portfolios.

5.2 Correlation Structure of Asset Returns

This section examines how diversification depends on the movements of assets, with respect to each other. This analysis helps identify co-movement patterns.

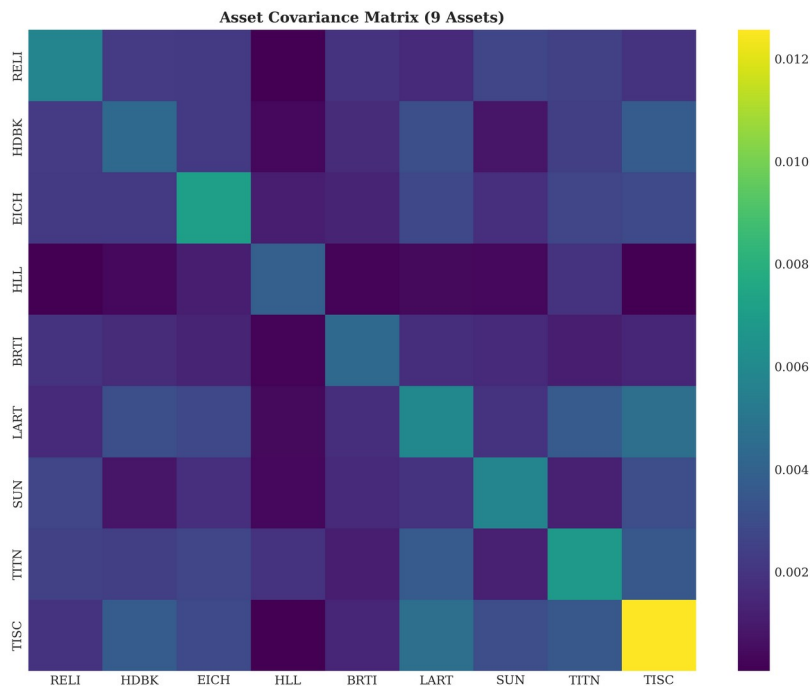


Chart 5.2 (a)

As visible in *chart 5.2 (a)*, covariance measures joint variability of asset returns. The chart shows the variance of each stock. Higher diagonal values mean higher variance. The first group (RELI, HDBK, EICH, HLL) shows moderate and consistent numbers across the board, suggesting they behave similarly and have stable relationships with each other. The second group (LART, SUN, TITN, TISC) shows extreme values, either very high or very low, indicating they behave differently from the first group. SUN has the highest diagonal value (0.011), meaning it moves sharply on its own. This makes it the riskiest stock in the group. TITN (0.002) and TISC (0.0009) have the lowest diagonal values, meaning they hardly fluctuate. They are the safest, most steady stocks in the set. Most stocks have low to moderate variances (0.006 to 0.012), except SUN which is high. The stable stocks (TITN, TISC) offer safety but also show sudden spikes in covariance with each other, meaning they lose their stability when grouped together. Mixing stocks from the first group (like RELI, HDBK) with the stable ones (TITN, TISC) creates a balanced portfolio. Adding SUN increases risk significantly. The strong link among LART, SUN, TITN, and TISC means putting them together would limit risk reduction, they would all move in the same direction at once. This covariance matrix forms the foundation of mean-variance optimization.

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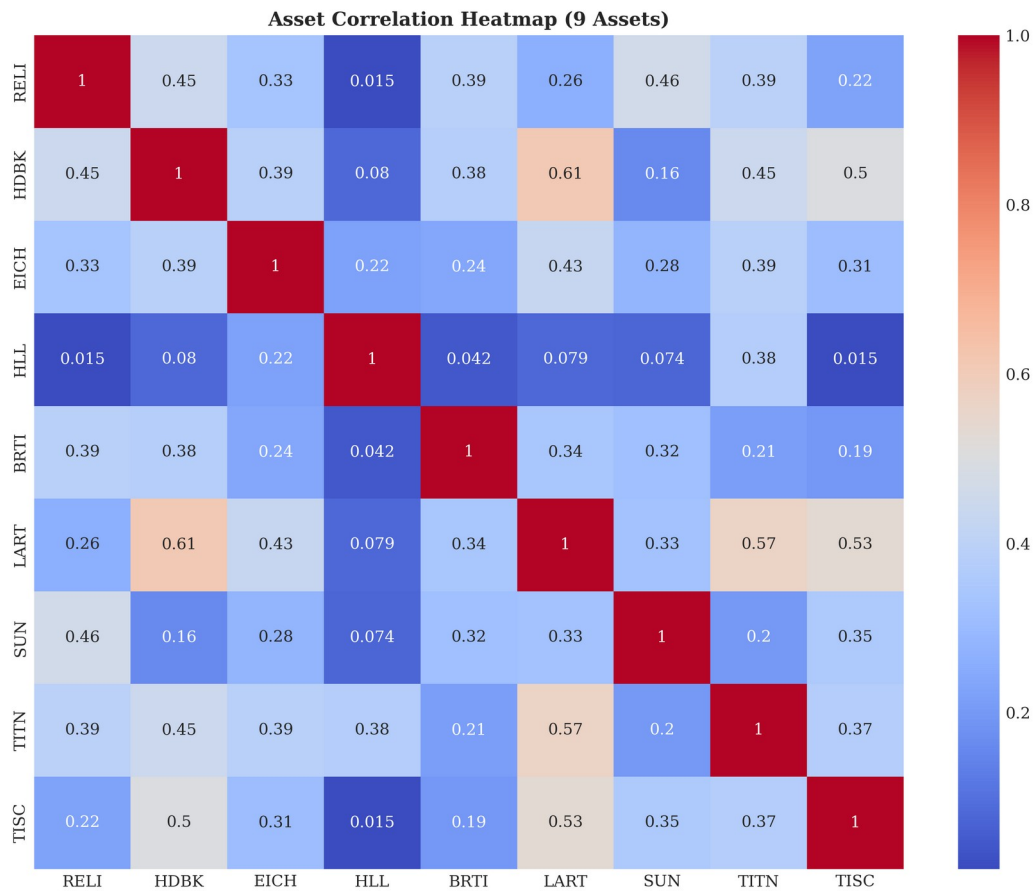


Chart 5.2 (b)

From *chart 5.2 (b)*, it is evident that HDBK and LART have the highest correlation at 0.61, meaning they move closely together. LART and TITN (0.57) and LART and TISC (0.53) also show strong positive relationships. HDBK is also highly correlated with TISC (0.50) and TITN (0.45). RELI and SUN are closely linked at 0.46. HLL is almost uncorrelated with several stocks: with RELI (0.015), TISC (0.015), BRTI (0.042), and LART (0.079). This means HLL moves independently of them. SUN and HDBK have a low correlation of 0.16, indicating they don't move much together. Many pairs fall in the 0.2 to 0.4 range, suggesting mild positive relationships. For example, EICH with HDBK (0.39) and RELI (0.33), BRTI with RELI (0.39) and LART (0.34), and TITN with RELI (0.39). The overall pattern shows HDBK, LART, TITN, and TISC form a cluster of stocks that tend to move together, with correlations mostly above 0.45 among them. HLL stands out as an outlier, having very low correlations with most others, suggesting it behaves differently from the rest. RELI and SUN also show a notable link, but SUN has weaker ties to others like HDBK (0.16) and TITN (0.20). Most assets show moderate positive correlations. This reflects a shared exposure to overall market movements. Some assets show relatively lower correlations, improving diversification. A notable finding is that no strong negative correlations exist, which is

typical for equity-only portfolios. This implies diversification benefits arise from differences in correlation strength.

Assets with lower correlation to the rest of the portfolio improve efficiency, and thus, optimization models tend to favour such assets in allocation decisions. Correlation and covariance relationships are used to construct the efficient frontier.

5.3 Construction of Traditional Portfolios

Traditional portfolios act as baseline portfolio construction approaches. They use equal-weight allocation across all assets. This “simple” portfolio helps in the evaluation of the impact of broad diversification without optimization techniques.

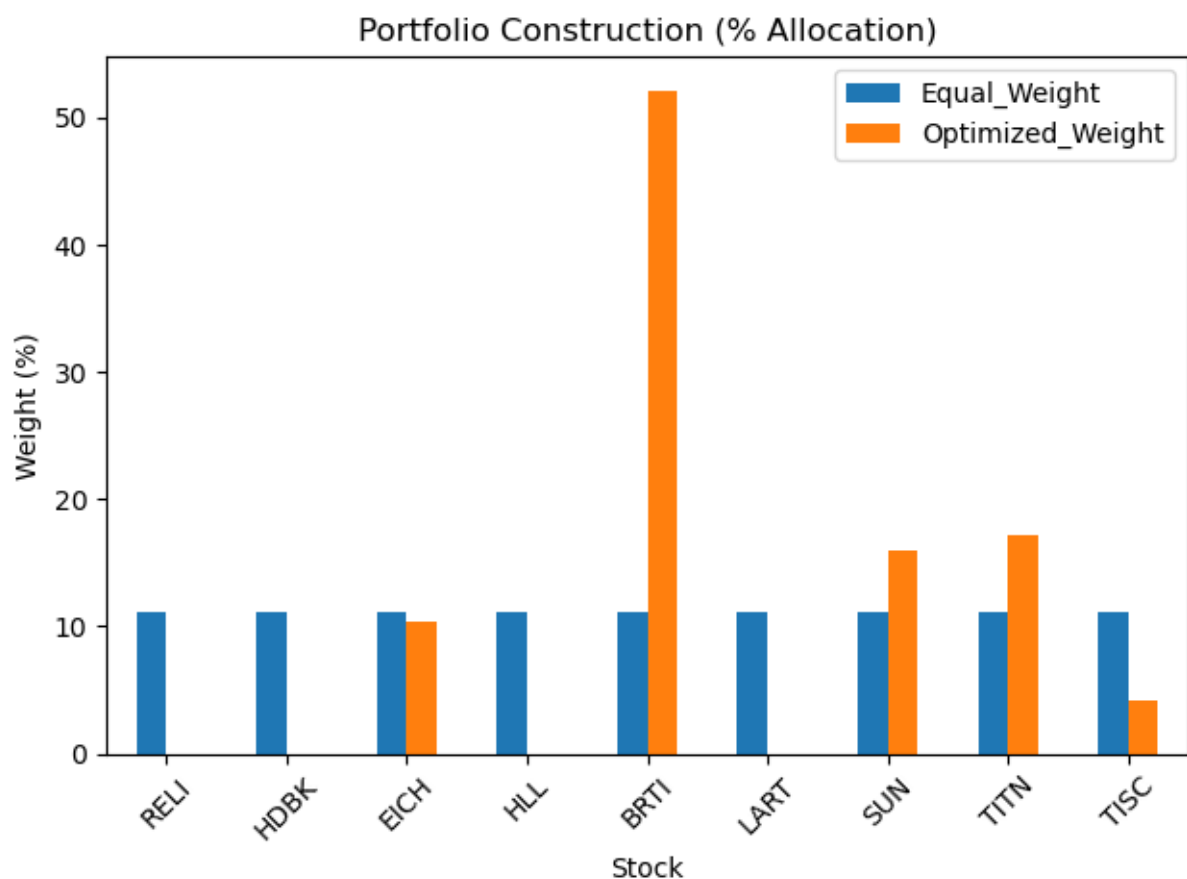


Chart 5.3 (a)

As shown in *chart 5.3 (a)* (and briefly discussed in section 4.4), each blue bar represents a uniform asset allocation. With nine assets, each asset is allocated 11.11% of the entire portfolio. Volatility, covariance, or expected returns is not considered. This portfolio employs simple diversification, i.e., portfolio weights remain static across assets. This strategy provides maximum diversification by number of holdings, which avoids concentrating risk in a single asset. The portfolio shows

simplicity, rather than mathematical optimization. This method is simple and transparent, allowing for easy implementation and rebalancing, and avoids subjective return forecasting. This strategy has limitations too. This method ignores correlation, and volatility differences. Such portfolios may produce suboptimal risk-return tradeoffs. They do not guarantee portfolio efficiency.

While equal weighting provides a useful benchmark, modern portfolio theory allows for more efficient allocation of capital. The next section constructs an optimized portfolio using mean–variance optimization and the efficient frontier.

5.4 Construction of Optimized Portfolios Using Mean–Variance Framework

This section deals with the application of the Markowitz mean–variance optimization to the stock basket (*table 4.4*), to determine asset weights, and maximise risk adjusted returns, by using historical returns of each asset, variance, and covariance estimates as inputs. As briefly mentioned in section 4.5, and shown in *chart 5.2 (a)*, portfolio weights are determined by expected returns, and covariance matrices, where the total summation of asset weights is equal to one, and no short selling is “allowed” (the reason is mentioned in section 4.4). These conditions are called “constraints”. The model (which can be found [here](#), and in the references) evaluates a significant number of portfolio combinations, to produce the “best portfolio”, i.e., the portfolio with the most optimal risk-reward(or return) trade-off.

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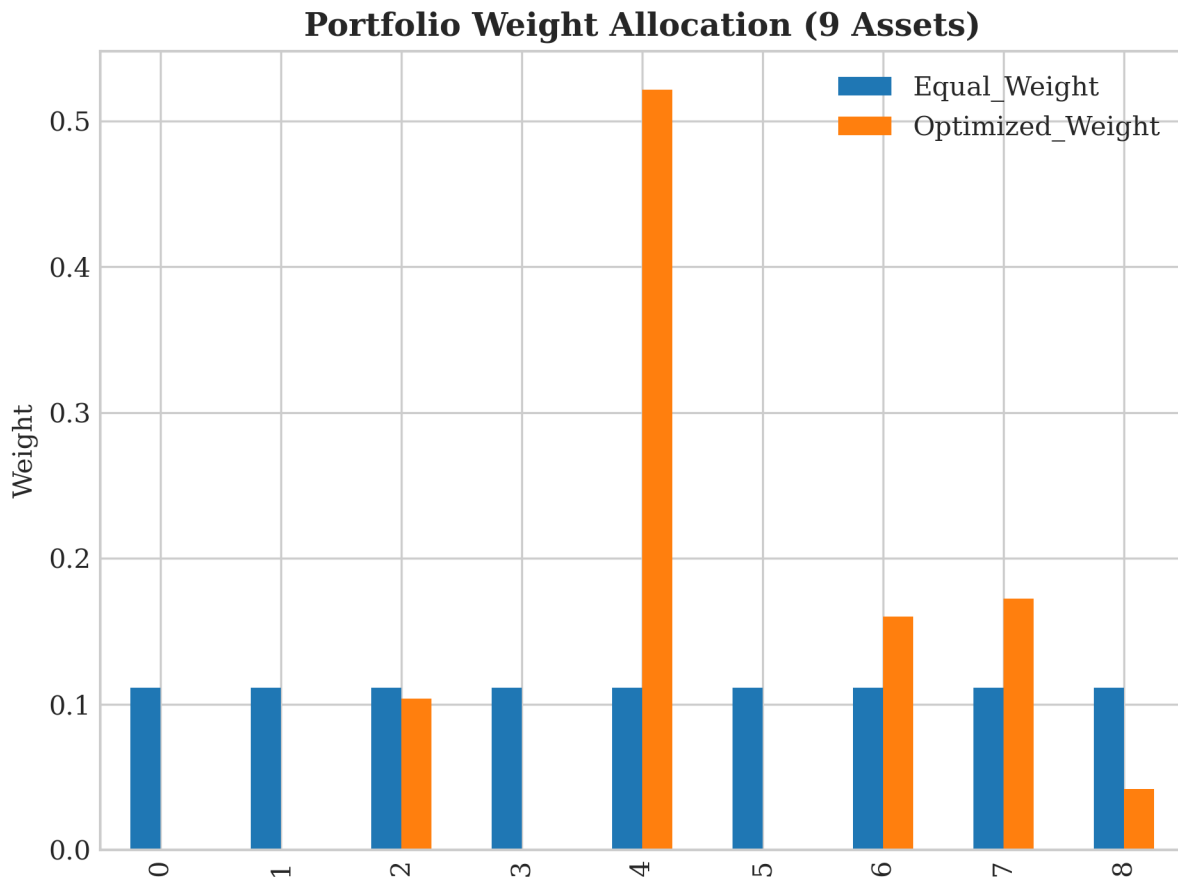


Chart 5.4 (a)

As visible in *chart 5.4 (a)*, allocation across assets becomes uneven (unlike the traditional approach). From *chart 5.3 (a)*, and *table 4.4*, it is evident that there is a great variation in allocations by the model. Bharti Airtel (BRTI) receives a dominant allocation (~52%), while several assets like RELI and HDBK get zero weight. This implies that the optimizer prioritises assets with strong diversification benefits. On the portfolio construction front, these results show that asset selection in modern portfolios is driven by risk contribution, and correlation structures. Optimization reduces exposure to redundant or highly correlated assets. Thus, the resulting portfolio represents maximum Sharpe ratio allocation.

5.5 Efficient Frontier Analysis

The purpose of this section is to analyse the efficient frontier, and examine risk-reward profiles across feasible portfolios. Some portfolios may offer relatively high expected returns but expose investors to greater volatility, while others may provide more stability at the cost of lower returns. By comparing these portfolios collectively, it becomes possible to distinguish between portfolios that are efficient and those that are dominated by better alternatives. The efficient frontier, therefore, serves as a visual and analytical representation of the most desirable trade-offs available to investors within the defined set of assets.

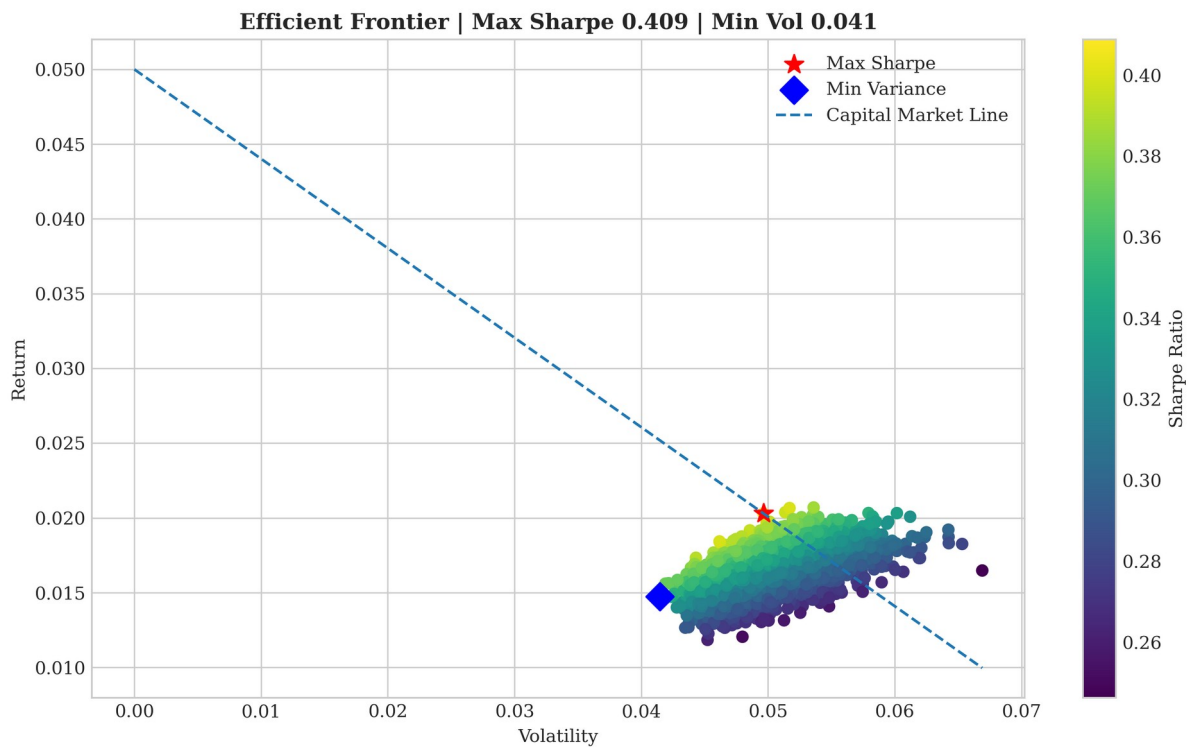


chart 5.5 (a)

Chart 5.5 (a) helps visualise the concept of the efficient frontier. The multiple points (which appear like a gradient “blob” (or dense cluster)), represent simulated portfolios (recall the analogy presented in section 1. If we extend that analogy to the modern approach of portfolio construction, we create thousands of fruit baskets, to find the right balance between a sweet and sour taste, here arbitrarily, sweet and sour can be understood as reward and risk!) . Each point is a portfolio. The capital market line represents portfolios that combine a risk-free asset with a risky portfolio on the frontier. The CML starts at the risk-free rate on the Y-axis and touches (“is tangent to”) the Efficient Frontier at the Max Sharpe portfolio. Portfolios on this line to the right of the tangency point represent leveraged positions (borrowing at the risk-free rate to invest more in the risky portfolio). The maximum Sharpe ratio of 0.409 refers to the optimal portfolio that provides the highest return per unit of risk (the tangency portfolio). The minimum volatility value of 0.041 refers to the portfolio with the absolute lowest risk (volatility) on the entire frontier. The colorbar ranging from “Min Sharpe Ratio” to “Max Sharpe Ratio” acts as a heat map. As one moves along the frontier from the “Min Variance” point towards the “Max Sharpe” point, the color shifts, indicating that you are improving the efficiency of your return per unit of risk, even as total volatility increases. The efficient frontier, therefore, illustrates how diversification across assets with varying correlations allows investors to achieve superior risk–reward combinations.

5.6 Comparative Risk–Return Analysis of Portfolios

This section deals with comparing the equal-weighted (traditional) portfolio with the optimized portfolio constructed using modern portfolio techniques. Such a comparison allows for the evaluation of the risk-adjusted performance of both strategies and ultimately helps determine the more optimal portfolio construction approach.

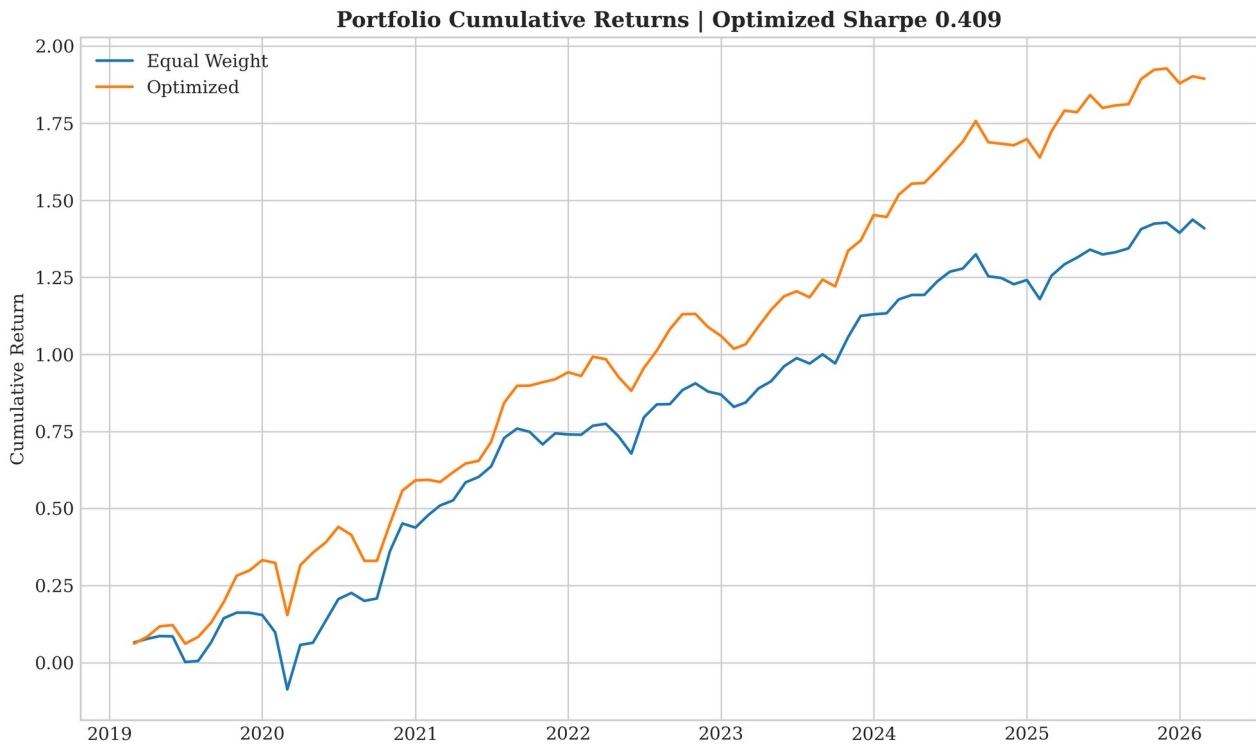


Chart 5.6 (a) [same as chart 4.6 (b)]

Chart 5.6(a) highlights the cumulative returns generated by both strategies over the sample period. It is evident that the optimized portfolio demonstrates stronger cumulative growth when compared to the equal-weighted arrangement. While both portfolios exhibit upward movement over time, the optimized portfolio maintains a consistently stronger trajectory, particularly in the latter half of the sample period. Another notable observation is the point at which the performance of the two portfolios begins to diverge significantly. This divergence illustrates the limitations of “blind” broad-based diversification through simple equal weighting, when compared to the systematic application of portfolio optimization techniques. The optimized strategy allocates capital based on the interaction of expected returns and covariance structures, allowing it to better manage risk while enhancing potential returns. A further point of comparison lies in the Sharpe ratios of the two portfolios. The optimized portfolio records a Sharpe ratio of approximately 0.4, whereas the equal-weighted portfolio exhibits a Sharpe ratio closer to 0.3 (Table 4.6(a)). This difference indicates that the optimized portfolio delivers higher returns per unit of risk, thereby demonstrating superior risk-adjusted performance.

Taken together, these results suggest that portfolio optimization can meaningfully improve investment outcomes by enhancing risk-adjusted returns. They also highlight the critical role played

by portfolio construction methods in determining overall performance. In this context, the findings provide empirical support for the advantages proposed by Modern Portfolio Theory.

It is important to note, however, that this study is conducted within a highly controlled and theoretical framework. The results primarily illustrate the conceptual strengths of portfolio optimization rather than its complete real-world applicability. Modern Portfolio Theory, like any financial model, has several practical limitations (as discussed in section 7.2).

Section 6: Empirical Results and Hypothesis Testing

The previous sections (4 and 5) dealt with finding results validating the viability of a portfolio built on the Modern Portfolio Theory, with the purpose of creating the most optimal portfolio which strikes the perfect balance between risk and reward (or return), versus a traditional approach of “blind” diversification. This section sets out to validate hypotheses (H1-H5).

H1: Active and passive portfolio construction strategies differ in their impact on portfolio returns.

The optimized (or active) portfolio demonstrated higher cumulative returns, delivering 20% compared to the equal-weighted (or passive) portfolio, which delivered ~15% (*chart 5.6(a)*). Risk-adjusted performance also improved, with Sharpe ratio increasing from 0.335 (passive) to 0.409 (active).

Result: Hypothesis validated. Portfolio construction strategies significantly influence portfolio returns.

H2: The level of diversification and asset correlation significantly influence portfolio risk, measured through standard deviation and beta.

Correlation analysis (*chart 5.2 (b)*) revealed clusters of highly correlated assets (e.g., HDBK–LART, LART–TITN) and several low-correlation relationships (e.g., HLL with most assets). This showed assets with lower correlations contributed to improved diversification and lower portfolio volatility. The optimized portfolio also exhibited a lower beta (0.91) compared to the equal-weighted portfolio (1.01), shown in *table 4.6(a)*.

Result: Hypothesis validated. Diversification structure and asset correlations significantly influence portfolio risk.

H3: Risk factors including systematic and unsystematic risk significantly affect risk adjusted returns, measured through Sharpe Ratio and Jensen's Alpha.

Risk-adjusted performance metrics differed substantially between the two portfolios. The optimized portfolio recorded higher Sharpe ratio (0.409), higher Treynor ratio (0.024 vs 0.016), and positive Jensen's Alpha (0.007), whereas the equal-weighted portfolio showed near-zero or slightly negative alpha (table 4.6 (a)).

Result: Hypothesis validated. Systematic and unsystematic risk factors significantly affect risk-adjusted returns.

H4: The use of optimization techniques such as mean variance analysis leads to higher portfolio efficiency and improved positioning on the efficient frontier.

Mean-variance optimization generated a portfolio located on the efficient frontier, maximizing the Sharpe ratio under the defined constraints. Asset allocation differed significantly from equal weighting, concentrating capital in assets with favorable return-risk and correlation characteristics (e.g., high allocation to BRTI).

Result: Hypothesis validated. Optimization techniques improve portfolio efficiency.

H5: Modern portfolio construction methods combined with active or passive strategies are expected to generate superior risk adjusted performance compared to traditional approaches.

The optimized (modern) portfolio consistently outperformed the traditional equal-weighted portfolio in both absolute returns and risk-adjusted metrics. The improved Sharpe ratio, Treynor ratio, Jensen's Alpha, and cumulative performance indicate superior efficiency of the modern approach.

Result: Hypothesis validated. Modern portfolio construction combined with systematic allocation methods produces superior risk-adjusted performance.

Overall Findings:

The research evidence strongly suggests that portfolio methodology plays a very important role in investment outcomes (or "rewards"). Modern optimization techniques, building on mathematical frameworks provide significant advantages over the traditional [equal-weight] diversification approach.

Section 7: Discussion

7.1 Empirical Support for Modern Portfolio Theory

This work supports the Modern Portfolio Theory. The study finds the optimized portfolio, which employed “strategic” asset allocation, using mean-variance, to have a higher Sharpe, compared to the traditional portfolio, which employed a simplistic equal-weighted broad allocation methodology. Another highlight is the higher cumulative returns delivered by the optimized portfolio, despite a lower beta compared to the traditional portfolio. The portfolio also managed to deliver better risk-adjusted returns, compared to the traditional portfolio. These findings provide empirical support for the efficient frontier and the core propositions of Modern Portfolio Theory. These metrics highlight the differences between the strategies of the modern approach to portfolio building, and the traditional approach. The modern approach prioritises allocation based on risk minimisation, and (or) reward maximisation, through prioritising “efficiency”, in contrast to the traditional equal-weighted solution, which intends to achieve the same outcome by maximising exposure, making this approach less nuanced than the modern approach. The modern “style” employs mathematical modelling, where weights are determined by expected returns and covariance, allocating capital to more efficient assets. Instead of focusing on why the modern portfolio fared better, understanding why the traditional portfolio “lagged” behind would be more “explorative” (in the words of Charlie T. Munger, “Invert, always invert!”). The broad based allocation allowed deployment of capital towards low-performing, and (or) high volatility assets, making it far less efficient than the modern portfolio, which avoided equal exposure to low-performing or highly volatile assets, and hence showed allocation discipline. The role of diversification and correlation warrant deeper expounding.

A portfolio is as good as its constituents, and how they interact with each other. In formal terms, portfolio risk depends not only on the variance of individual assets but also on the covariance structure between them. Portfolio risk (and performance), hence, depends on interaction between assets (or “constituents”). Combining assets with low or imperfect correlation reduces portfolio volatility without proportionally reducing expected return.

The traditional approach ignores intra-portfolio relationships, making it more prone to volatility. In the optimized method, diversification becomes strategic, rather than random, avoiding imperfect inter-asset correlation, and [therefore] maximising the benefits of diversification.

This study reveals several practical implications for investment strategies. First, portfolio design data should be data driven, because data helps curtail even the slightest “feeling” of entropy, by making the entire process systematic. Second, investors (and their investment strategies) must take into account risk-adjusted thinking. Third, investors (and their portfolios) benefit from structured asset allocation. Fourth, passive investors may still benefit from optimization frameworks. Though this work critiques a “pure passive” approach, a hybrid approach which mixes the “best of both worlds” would certainly be of benefit. This work also provides several strategic insights for portfolio managers. By treating portfolio construction as a scientific procedure, rather than an arbitrary process, and by diversifying with intent, along with strategic allocation driven by structured frameworks, long-term institutional portfolio management, anchored with robust risk management, can benefit greatly.

7.2 Critical Perspectives and Study Limitations

The Modern Portfolio Theory, with its allure of certainty, driven by the use of “pretty mathematics” and “comfortable” assumptions, often gives a false sense of “knowing”, which proves to be a very dangerous thing, when it comes to the creation of capital. The Modern Portfolio Theory banks on some “absurd” assumptions. The first assumption, in this saga of absurdities, is the assumption of rationality. This theory assumes perfectly rational investors, which is simply untrue. Humans, with all their wits and logic, in the end, are irrational. The advent of algorithmic trading has pushed the efficiency of the market to higher levels, say, from a few decades ago, but irrationality persists. By ignoring behavioural biases, utility maximisation, and the “emotion” of the market, the MPT distances itself from the realms of reality. From ignoring key real-market conditions, like transaction costs, taxes, borrowing rational, and assuming perfect liquidity the MPT undermines its practical applicability. Perhaps, the greatest drawback of the MPT is that it is purely quantitative, rather than being a mix of a qualitative, and quantitative approach.

The most potent evidence of this is assuming risk equals volatility, which in the “long-run”, is untrue. A more qualitative (and “logical”) definition of risk is the potential loss of capital invested, whereas, volatility can be defined, simply, as fluctuations in the price of an asset. Equating these two, seems to be like comparing apples to oranges. These points led to alternate approaches, like Merton’s inter-temporal models.

This study has several limitations too, which warrant addressing. First, and perhaps the most glaring “flaw”, is setting the risk-free rate a 5% p.a., as opposed to the usual 6.5% to 7.5%. This was done for the purposes of modelling. This subtle difference “balloons” to huge differences in the ratios calculated, but the underlying academic revelations remain unchanged. This also highlights another limitation of this study, and that is the model.

The model employed for this study, just like any other model, is highly sensitive, influenced greatly by assumptions, making it less reliable for practical implementations. The third limitation, is creating a synthetic market. These kinds of markets create circularity. The “market” is the investable universe. Any portfolio constructed from these nine stocks will naturally be highly correlated with this synthetic benchmark. This was done because the portfolio universe is limited to nine securities, using NIFTY-50 would introduce structural mismatch between the benchmark and the sample. Therefore a synthetic equally weighted market proxy was constructed. The fourth limitation is the assumption of frictionless trading. This study does not take into account rebalancing and trading costs, making it more academic than practical. Another limitation of this study is the assumption that all passive portfolios employ equal-weighted strategies, which is not always the case. This was done to simplify the research methodology, and focus on the academic contrasts between portfolio management “styles” (or strategies).

Overall, the greatest limitation of this study, and the MPT is that these are vastly theoretical. From unrealistic assumptions to sensitivities, these works lack real world applicability. Though this study, and MPT preserved academic rigour, they, somewhere along the way, lost practical “realism”.

Section 8: Conclusion and Recommendations

This work was a comparative study which contrasted traditional equal-weight portfolio construction strategies against optimized portfolio construction built on Markowitz's modern portfolio theory. This research spanned three distinct phases witnessed by the Indian capital markets, covering the time period between 2019 and early 2026. Testing across this period revealed that the optimized portfolio achieved superior risk-adjusted performance relative to the equal-weighted benchmark. The optimized portfolio showed a higher Sharpe ratio (0.409 vs 0.335(traditional portfolio)), indicating stronger returns per unit of total risk. The optimized portfolio also showed lower systematic risk ($\beta \approx 0.91$) compared with the equal-weighted portfolio ($\beta \approx 1.01$). Treynor ratio and Jensen's Alpha further confirmed the outperformance of the optimized portfolio, suggesting better compensation per unit of systematic risk. Cumulative return analysis indicated that the optimized strategy generated roughly 20% cumulative growth, outperforming the equal-weighted portfolio by approximately 500 basis points. Correlation and covariance analysis showed that diversification benefits depend on the interaction between assets, not simply the number of assets held. The findings provide empirical support for the core propositions of Modern Portfolio Theory, particularly the role of diversification and the efficient frontier.

These results lead to some important implications for portfolio management. First, portfolio construction significantly influences investment outcomes, even when the underlying asset universe remains constant. Second, strategic diversification based on correlations and risk contribution is more effective than simple broad diversification. Third, quantitative optimization techniques can improve capital allocation efficiency by identifying assets that enhance risk-adjusted performance. The final, and perhaps the most "practical" implication is that even passive investment approaches may benefit from periodic optimization frameworks to improve portfolio efficiency.

The implications, hence, lead to actionable recommendations for investors and portfolio managers. First, portfolio construction should incorporate correlation analysis and diversification structure, rather than relying solely on equal allocation. Second, portfolio managers should consider optimization tools based on Modern Portfolio Theory when designing diversified portfolios. Third, and the most pivotal, portfolio design should integrate both quantitative modelling and qualitative judgment to account for market structure and macroeconomic conditions.

Suggestions for Future Research

A major limitation of this study was the small asset universe, future research could expand this universe to include multiple asset classes, sectors, and instruments. Future works could incorporate market benchmarks such as the NIFTY 50 or BSE 500 to further improve CAPM-based performance evaluation. Finally, long-horizon studies covering multiple market cycles could provide deeper insights into the robustness of optimized portfolios across varying economic environments.

Section 9: Glossary

Beta (β)

A measure of systematic risk that indicates how sensitive an asset or portfolio's returns are to movements in the overall market.

Capital Asset Pricing Model (CAPM)

An equilibrium asset-pricing model that describes the relationship between expected return and systematic risk, measured by beta.

Correlation

A statistical measure ranging from -1 to $+1$ that indicates the strength and direction of the relationship between the returns of two assets.

Covariance

A statistical measure that indicates the degree to which the returns of two assets move together.

Efficient Frontier

The set of optimal portfolios that offer the highest expected return for each level of risk within a given set of assets.

Jensen's Alpha

A performance metric that measures the excess return of a portfolio relative to the return predicted by the CAPM.

Mean-Variance Optimization

A portfolio construction method introduced by Harry Markowitz that determines asset allocations by maximizing expected return for a given level of risk or minimizing risk for a given expected return.

Portfolio Diversification

The practice of allocating investments across different assets or sectors to reduce overall portfolio risk.

Sharpe Ratio

A risk-adjusted performance measure that evaluates excess return relative to the risk-free rate per unit of total portfolio volatility.

Systematic Risk

Market-wide risk arising from macroeconomic factors that cannot be eliminated through diversification.

Treynor Ratio

A risk-adjusted return metric that measures excess return earned per unit of systematic risk (beta).

Unsystematic Risk

Asset-specific risk that can be reduced or eliminated through diversification.

Section 10: References

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Section 11: Appendices

Appendix A: Stock Universe Used in the Study

Company Name	NSE Symbol	Name Used in Study	Sector
Reliance Industries Ltd.	RELIANCE	RELI	Energy / Conglomerate
HDFC Bank Ltd.	HDFCBANK	HDBK	Banking
Eicher Motors Ltd.	EICHERMOT	EICH	Automobile
Hindustan Unilever Ltd.	HINDUNILVR	HLL	FMCG
Bharti Airtel Ltd.	BHARTIARTL	BRTI	Telecom
Larsen & Toubro Ltd.	LT	LART	Infrastructure / Engineering
Sun Pharmaceutical Industries Ltd.	SUNPHARMA	SUN	Pharmaceuticals
Titan Company Ltd.	TITAN	TITN	Consumer Discretionary
Tata Steel Ltd.	TATASTEEL	TISC	Metals

Table A1: List of companies included in the study along with their NSE symbols and sector classifications.

Appendix B: Raw Data Structure

This appendix presents the structure of the dataset used in the study.

Variable	Description
Date	Trading day
Price	Closing price
Open	Opening price
High	Highest price during the trading day
Low	Lowest price during the trading day
Volume	Number of shares traded
Change %	Daily percentage change in price

Table B1: For the purposes of the analysis, Date and Change % were used to compute monthly returns

